

Portfolio Question 1.2.P - Metrics

Daniel Lawson University of Bristol

Portfolio 1.2.P (v4.0.0)

Portfolio Example Question

- ▶ Portfolio Question 1.2.P: When is an accurate classifier not useful? Can we ensure it is useful?
 - ▶ What metrics are better than accuracy? In what sense are they reliable, and do they relate to uncertainty?
 - ▶ Hints: take a look at Proper Scoring Rules
- ▶ Format
 - ▶ Introduction
 - ▶ Non-trivial implication
 - ▶ Discussion

Context

- ▶ We were introduced to ROC (Receiver-Operating Curves) in Lecture 01.2
- ▶ Definition:
 - ▶ **Accuracy** is the proportion of correctly classified points.
 - ▶ But... what is the **baseline**? If there are imbalanced classes?
- ▶ If we optimise for a metric with a given test/train setup:
 - ▶ Our model will perform well for that setup
- ▶ Can we say anything more?
 - ▶ Can we reflect **real-world** use?

Why is Accuracy bad?

- ▶ Accuracy is a natural metric
- ▶ It goes wrong if classes are imbalanced, e.g.
 - ▶ A 99-1 split between classes A and B
 - ▶ Always guess A
 - ▶ Gets 99% accuracy
- ▶ It also ignores uncertainty
 - ▶ May reward overly-certain claims more than calibrated uncertainty

Metrics for Classification

- ▶ Start with the ROC curve:
 - ▶ TPR **at a given FPR**
 - ▶ AUC characterises the whole ROC curve
- ▶ Area Under Precision-Recall Curve (**AUPRC**) is also a thing people advocate for
- ▶ None are “right”, we have to define the inference task
- ▶ Any of these and more are often optimized
 - ▶ If we optimise a parameter or perform model comparison based on test data, we need additional test data to test the meta-algorithm!

PR curve

- ▶ Nice things¹ about ROC and PR curves:
- ▶ Dominance:
 - ▶ If one curve dominates (is always above) another in ROC, it dominates in PR
 - ▶ and vice-versa
- ▶ ROC curves can be **linearly interpolated**
 - ▶ This is “flipping a coin” to access classifiers in-between
- ▶ PR curves have a slightly more complex relationship but the same principle can be applied
- ▶ Integrating both scores leads to performance metric that can be optimized

¹Davis and Goadrich, “The Relationship Between Precision-Recall and ROC Curves”, ICML 2006.

Non-trivial point: towards Proper Scoring Rules

- ▶ Use $P(Y|X)$ as the loss:
- ▶ 'A scoring rule is proper if the forecaster maximizes the expected score for an observation drawn from the distribution F if he or she issues the probabilistic forecast F , rather than $G \neq F$ proper scoring rules encourage the forecaster to make careful assessments and to be honest.'
 - ▶ Gneiting and Raftery 2007.
- ▶ Easy [wikipedia article](#)
- ▶ Examples include: Log-likelihood, Brier score
$$\sum_{i=1}^n (y_i - \hat{p}_i(\mathbf{x}_i))^2$$
- ▶ Takehome: We need to keep probabilities in the model

Discussion

- ▶ What do we want from a loss function?
- ▶ What is the relationship between a loss function and a model?
- ▶ Does finding parameters to minimise the loss mean that the model is right?
 - ▶ How does a proper score rule work on the logistic regression example in class?
- ▶ Is this the same as calibration?
- ▶ When should we care?
 - ▶ Is this advice still up to date?
 - ▶ Do e.g. large language models care about this?
 - ▶ How could it work for imaging?